



Scale–Clock Geometry

A Mathematical Foundation for RTM

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Abstract

We develop a rigorous foundation for **Multiscale Temporal Relativity (RTM)** as a mathematical theory of **time-scale scaling**. Starting from a scale-semigroup axiom $T(bL) = f(b)T(L)$ with mild regularity, we derive the **power law** $T(L) = \kappa L^\alpha$, identifying the **coherence exponent** α as a **clock-invariant** slope and κ as a gauge (clock) factor. We recast RTM in geometric form via the 1-form

$$\omega = d(\log T) - \alpha d(\log L),$$

and prove that **collapse**, residual independence of $\log T - \alpha \log L$ from $\log L$ is equivalent to **exactness/flatness** of ω on a bin; regime mixing and non-power alternatives appear as **holonomy/curvature**. We embed RTM in **regular variation** with **variable exponents**, quantifying finite-window bias and showing that collapse statistics scale with curvature. A **renormalization** operator (scale dilation + re-gauge) has power laws as **fixed points** and is **contractive** in Hölder/Zygmund classes; slowly varying clocks lie on a **center manifold**, and slowly drifting exponents yield **adiabatic attraction**. In dynamics, RTM acts as a **space-dependent clock** for diffusions and Dirichlet forms, giving similarity exponents $z = m + \alpha$ and exit-time laws $T \sim R^{m+\alpha}$ with adiabatic error bounds. For inference under errors-in-variables, we show consistency of **ODR/TLS**, **SIMEX**, and **Theil-Sen** for local α , and formalize the collapse statistic as a specification test against curvature. A category-theoretic packaging makes clocks a gauge and slope the moduli invariant, clarifying functorial behavior under products and coarse-graining. We conclude with constructive counterexamples and open problems (holonomy tests, graph settings, inverse problems, heavy-tailed noise).

1. Introduction

1.1 Problem and viewpoint

Many systems exhibit a systematic relation between a **characteristic time** T and a **scale proxy** L : larger units operate on slower clocks, smaller units on faster clocks. RTM posits that **inside a fixed environment** this relation is **multiplicatively consistent** under rescaling of L . The empirical practice, seen across physics, biology, and economics, is to examine the slope of $\log T$ vs. $\log L$ and test whether residuals “collapse” after detrending by that slope.

This paper supplies a **mathematical backbone** for that practice. Our central claim is that the **slope** α is the structural object (invariant under clock changes), while **clocks** are a gauge. With this separation, RTM becomes a clean theory linking: (i) functional equations $\rightarrow$ power laws, (ii) a **1-form/connection** whose flatness encodes collapse, (iii) **regular variation** with variable exponents to quantify finite-window effects, (iv) **renormalization** as scale-dilation dynamics with power-law fixed points, (v) **diffusions with space-dependent clocks**, and (vi) **statistical identifiability** under measurement error.

Scope note. This document operates under Assumptions 1–6 of Doc 001 (Sec. 2.1): local density ρ and temperature θ are uniform within each bin. Under these conditions, the full master law $T/T_0 = (L/L_0)^\alpha \cdot \theta(\mathcal{T})/\sqrt{\rho/\rho_0}$ reduces to $T = \kappa L^\alpha$. For treatment of variable ρ and θ , see Doc 001 Sec. 2.1–2.2. For empirical applications where these assumptions may be violated, the collapse test (Sec. 3.2, 7) provides a falsifiable diagnostic.

1.2 Contributions

1. **Semigroup $\rightarrow$ Power law (Sec. 2).**
From $T(bL) = f(b) T(L)$ with measurable/continuous f , we obtain $f(b) = b^\alpha$ and $T(L) = \kappa L^\alpha$. The slope α is **clock-invariant**; intercept $\log \kappa$ is the clock.
2. **Scale-clock geometry (Sec. 3).**
With $\omega = d \log T - \alpha d \log L$, we prove **collapse $\Leftrightarrow$ exactness/flatness** of ω on a bin. Regime mixing and non-power behavior manifest as **holonomy** ($\oint \omega \neq 0$). We quantify **adiabatic collapse** when α drifts slowly.
3. **Variable-exponent regular variation (Sec. 4).**
We formalize $T(L; x) = L^{\alpha(x)} \ell(L; x)$ with **uniform** slow variation, derive **finite-window bias** $O(\|\partial_u \alpha\| h)$, and show collapse statistics scale like curvature $\sim h^2$.
4. **Renormalization & stability (Sec. 5).**
A dilation-plus-re-gauge operator has power laws as **fixed points**; in Hölder/Zygmund classes it is a **contraction**, giving **local attraction** to the power-law manifold. Slowly varying clocks form a **center manifold**; slowly drifting exponents yield **adiabatic tracking**.
5. **RTM diffusions & Dirichlet forms (Sec. 6).**
Let conductivity be $L(x)^{-\alpha(x)}$. With α constant, solutions obey self-similarity with dynamic exponent $z = m + \alpha$ and exit-time scaling $T \sim R^z$; with slow drift we obtain **adiabatic error bounds**. RTM diffusions are **time-changed Brownian motions**.
6. **Identifiability & inference under EIV (Sec. 7).**
We show **ODR/TLS** and **SIMEX** consistently recover local α under standard assumptions; **Theil-Sen** provides robust checks. The **collapse statistic** is a specification test against curvature even with measurement error.
7. **Categorical packaging (Sec. 8).**
Objects carry ω and a gauge; **slope** is the moduli invariant, clocks are gauge; products add slopes; coarse-graining is a functor with power-law fixed points.
8. **Examples, counterexamples, open problems (Sec. 9).**
We give constructive failures (kinks, curved log-log), and list open directions (holonomy tests, graphs, inverse problems, heavy-tailed errors).

1.3 Relation to prior work

Our use of multiplicative Cauchy equations and **regular variation** follows Karamata/de Haan, but adapted to **variable exponents** uniform in environment. The 1-form framing parallels standard **gauge/connection** ideas, here specialized to a scale-clock bundle so that “collapse” is **exactness**. The renormalization operator is classical in spirit but applied to **log-time functions** with **gauge equivalence**; the contraction on Hölder/Zygmund spaces provides a clean path from empirical scaling to a dynamical fixed-point picture. The diffusion results connect to **time-changed processes** and variable-coefficient elliptic theory, giving explicit similarity exponents tied to α . On inference, our statements place **errors-in-variables** estimators inside the RTM invariance structure, clarifying what is and isn’t identifiable.

1.4 Scope and falsifiability

RTM is intended for **bins**, domains where environment is stable enough that clocks are L independent. The theory **predicts its own failure modes**: non-power curvature or regime

mixtures produce holonomy and non-vanishing collapse statistics. These are **scope boundaries**, not defects.

1.5 Paper roadmap

- **Sec. 2:** semigroup axiom $\Rightarrow$ power law; clock invariance.
- **Sec. 3:** scale–clock geometry; collapse = exactness; holonomy and adiabatic collapse.
- **Sec. 4:** variable-exponent regular variation; finite-window bias; curvature tests.
- **Sec. 5:** renormalization fixed points; contraction and center manifold; adiabatic tracking.
- **Sec. 6:** RTM diffusions/Dirichlet forms; similarity exponent $z = m + \alpha$; exit times; time change.
- **Sec. 7:** identifiability and consistency under EIV; collapse as specification test.
- **Sec. 8:** categorical formulation and functorial properties.
- **Sec. 9:** examples, counterexamples, open problems; concise conclusion.

Guiding principle: *structure lives in the slope; clocks live in the gauge*. The remainder of the paper makes this precise across analysis, geometry, dynamics, and inference.

2. Scale Semigroup $\rightarrow$ Power Law (Foundations)

This section formalizes the scaling axiom behind RTM and derives the power-law form $T(L) = \kappa L^\alpha$. We also isolate the **clock** as a multiplicative gauge and prove that **slope** α is the structural invariant. Throughout, $L \in \mathbb{R}_{>0}$ denotes a size/scale variable and $T(L) \in \mathbb{R}_{>0}$ a characteristic time.

2.1 Axioms and consequences

We separate **scale symmetry** from **clock choice**.

Axiom 2.1 (Scale semigroup).

There exists a family of maps $\{S_b\}_{b>0}$ (scalings by factor b) and a function $f: \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that for all $b > 0$ and all $L > 0$,

$$T(S_b L) = T(bL) = f(b) T(L),$$

with $f(1) = 1$ and

$$f(b_1 b_2) = f(b_1) f(b_2) \quad (\text{semigroup composition}).$$

Axiom 2.2 (Mild regularity).

Either (i) f is measurable on $\mathbb{R}_{>0}$, or (ii) f is continuous at $b = 1$.

(Any standard regularity —Baire/measurable/locally bounded— will do.)

Definition 2.3 (Clock transform).

A change of measurement units or baseline timing is a map $T \mapsto T^\#$ of the form $T^\#(L) = c T(L)$ for some constant $c > 0$ (or, more generally, $c = c(x)$ depending on an external **environment** parameter x , but *independent of* L within a fixed environment).

2.2 Functional-equation solution

Lemma 2.4 (Multiplicative Cauchy).

Under Axioms 2.1–2.2, $f(b) = b^\alpha$ for some $\alpha \in \mathbb{R}$.

Proof. Set $g(\log b) := \log f(b)$. The semigroup law gives $g(u + v) = g(u) + g(v)$. Measurability (or continuity at 0) forces $g(u) = \alpha u$ for some $\alpha \in \mathbb{R}$. Exponentiating, $f(b) = e^{g(\log b)} = b^\alpha$.

Theorem 2.5 (Power-law representation).

Fix any $L_0 > 0$. Under Axioms 2.1–2.2,

$$T(L) = T(L_0) \left(\frac{L}{L_0} \right)^\alpha = \kappa L^\alpha, \quad \text{where } \kappa := T(L_0) L_0^{-\alpha}.$$

Proof. Apply Lemma 2.4 with $b = L/L_0$:

$$T(L) = T((L/L_0)L_0) = f(L/L_0) T(L_0) = (L/L_0)^\alpha T(L_0).$$

Rearrange to define κ .

Corollary 2.6 (Log-linear form).

$\log T = \alpha \log L + \log \kappa$. Hence **slope** α captures scaling, while **intercept** $\log \kappa$ captures the clock.

2.3 Clock invariance and identifiability of α

Proposition 2.7 (Clock invariance of the slope).

If $T^\#(L) = c T(L)$ with $c > 0$ independent of L (within a fixed environment), then

$$\log T^\# = \alpha \log L + (\log \kappa + \log c),$$

so the regression slope of $\log T^\#$ on $\log L$ equals α .

Proof. Immediate from the corollary.

Remark 2.8 (Environment-dependent clocks).

If the clock factor depends on an external label x but not on L i.e. $T^\#(L; x) = c(x) T(L; x)$ then within any fixed x environment bin the slope stays $\alpha(x)$, while the intercept shifts by $\log c(x)$.

Proposition 2.9 (Uniqueness up to clock).

If $T_1(L) = \kappa_1 L^{\alpha_1}$ and $T_2(L) = \kappa_2 L^{\alpha_2}$ satisfy $T_2(L) = c T_1(L)$ for all L with some $c > 0$, then $\alpha_1 = \alpha_2$ and $c = \kappa_2 / \kappa_1$.

Proof. Take logs and compare coefficients of $\log L$.

2.4 Regular-variation generalization (optional but useful)

The exact scaling can be relaxed to **regular variation**, which covers slowly varying clocks and asymptotic power laws.

Definition 2.10 (Karamata regular variation).

A positive function T is *regularly varying of index* α if for all $b > 0$,

$$\lim_{L \rightarrow \infty} \frac{T(bL)}{T(L)} = b^\alpha.$$

Equivalently, $T(L) = L^\alpha \ell(L)$ with ℓ slowly varying ($\ell(bL)/\ell(L) \rightarrow 1$).

Theorem 2.11 (RTM under regular variation).

If T is regularly varying with index α , then local log–log slopes over compact $\log L$ windows converge to α . Changes of clock that are slowly varying (e.g., ℓ) perturb the intercept asymptotically but not the slope.

Sketch. Standard Karamata representation and Tauberian arguments; consistency of local slope follows from uniform convergence of ratios.

Remark 2.12 (Finite-window bias).

When ℓ is not flat on the observed range, $\hat{\alpha}$ is biased by $O(\sup |\log \ell|)$ across the window. This motivates **environment fixing** and narrow windows in empirical RTM.

2.5 Necessary and sufficient conditions for power-law scaling

The next statement packages RTM’s **collapse test** idea at the algebraic level.

Proposition 2.13 (Equivalence of power-law and log-affinity).

For a fixed environment, the following are equivalent:

1. $T(L) = \kappa L^\alpha$ for some $\kappa > 0, \alpha \in \mathbb{R}$.
2. There exist constants α, c such that $\log T - \alpha \log L \equiv c$ for all L .
3. For any $L_1 \neq L_2$,

$$\frac{\log T(L_2) - \log T(L_1)}{\log L_2 - \log L_1} \equiv \alpha \text{ (independent of the pair).}$$

Proof. (1) $\Rightarrow$ (2) is Cor. 2.6; (2) $\Rightarrow$ (3) by subtracting; (3) $\Rightarrow$ (2) by fixing L_1 and integrating constancy of the discrete derivative; then exponentiate to get (1).

Corollary 2.14 (Binwise specification test).

Given observations $\{(L_i, T_i)\}$ in a fixed environment, if a consistent slope α exists such that the residuals $\tilde{y}_i := \log T_i - \alpha \log L_i$ are **independent of $\log L_i$** (up to noise), the RTM power-law specification is not rejected for that bin. Any systematic trend of $\tilde{y}$ vs. $\log L$ falsifies exact power-law scaling in that bin.

2.6 Counterexamples and scope

Counterexample 2.15 (Mixtures of regimes).

Let $T(L) = \kappa_1 L^{\alpha_1}$ for $L \leq L^*$ and $T(L) = \kappa_2 L^{\alpha_2}$ for $L > L^*$ with $\alpha_1 \neq \alpha_2$. Then no single α fits globally; any attempt will show residual trend changes at L^* . (This is **regime mixing** and should be split into bins.)

Counterexample 2.16 (Curvature).

Let $\log T = g(\log L)$ with $g'' \neq 0$ on an interval. Then the discrete slope in Prop. 2.13 depends on the pair, violating the power-law condition; collapse must fail on that interval.

2.7 Finite-sample estimation with measurement error (set-up for later)

While proofs above are exact-function statements, empirical RTM faces noisy L, T . Write $x = \log L, y = \log T$, and observe

$$x^{obs} = x + \xi, y^{obs} = y + \zeta,$$

with mean-zero errors. Ordinary least squares attenuates the slope when $\xi \neq 0$. We will later show (Section 7) that **orthogonal distance regression** (total least squares) and **SIMEX** yield consistent $\hat{\alpha}$ under standard conditions; the invariance results (Props. 2.7–2.9) still hold exactly because clock multiplies T , not L .

2.8 Summary of Section 2

- The **scale semigroup** + mild regularity forces a **power law** $T = \kappa L^\alpha$.
- **Slope** α is **clock-invariant** and identifies the structure; **intercept** $\log \kappa$ encodes the clock.
- **Regular variation** extends the theory to asymptotic scaling with slowly varying clocks.
- RTM's **collapse** criterion is the algebraic statement that $\log T - \alpha \log L$ is constant (no trend vs. $\log L$) inside a bin.
- Regime mixtures and curvature provide clean **counterexamples**, justifying binning and specification tests.

3. Scale–Clock Geometry (Collapse as Exactness)

We give a geometric formulation of RTM that separates **slope** (structure) from **clock** (gauge). The key object is the 1-form

$$\omega = d(\log T) - \alpha(x) d(\log L),$$

defined on a product space where x indexes *environment* and $L > 0$ is *scale*. RTM's **collapse** criterion becomes the statement that ω is **exact/flat** on a bin. This section makes that precise and proves the equivalences.

3.1 Spaces, coordinates, and bins

- Let X be a smooth (or at least topological) **environment space** collecting background conditions (policy regime, technology, microstructure).
- Let $S = \mathbb{R}_{>0}$ be the **scale** line with coordinate L ; write $u = \log L \in \mathbb{R}$.
- Let $Y = \mathbb{R}_{>0}$ be the **clocked time** line with coordinate T ; write $v = \log T \in \mathbb{R}$.

We work on the manifold $M = X \times S$ with coordinates (x, u) . A **bin** is a path-connected open set $E \subset M$ on which “environment is fixed enough” in the RTM sense (no regime breaks). On E , assume a locally integrable **coherence field** $\alpha: E \rightarrow \mathbb{R}$.

3.2 The RTM 1-form and clock (gauge) transformations

Definition 3.1 (RTM 1-form).

On $E \subset M$, define

$$\omega = dv - \alpha(x, u) du.$$

Here α may depend on x and (optionally) on u if we allow slowly varying exponents; constant- α is the ideal RTM case.

Clock (gauge) transformations.

A **clock change** multiplies raw time by a positive factor independent of L inside the bin:

$$v \mapsto v^\# = v + \phi(x), \phi: X \rightarrow \mathbb{R}.$$

Under this,

$$\omega \mapsto \omega^\# = d(v + \phi(x)) - \alpha du = \omega + d\phi(x).$$

Thus ω is defined **up to addition of exact 1-forms pulled back from X** a standard gauge freedom.

Proposition 3.2 (Slope is gauge-invariant).

Clock changes $v \mapsto v + \phi(x)$ do not alter the α coefficient of du . Hence α is a gauge-invariant object, while v and ω shift by exact forms.

Proof. Immediate from the transformation rule.

3.3 Collapse as exactness

RTM's **collapse** states that, within a bin, after removing αu the remaining variation of v is constant (up to noise), i.e. residuals do not trend with u .

Theorem 3.3 (Collapse $\Leftrightarrow$ exactness).

Let $E \subset M$ be simply connected. The following are equivalent:

1. (*Power-law chart*) There exists a function $\kappa: E \rightarrow \mathbb{R}_{>0}$ such that

$$v(x, u) = \alpha(x)u + \log \kappa(x).$$

(Constant- α case; for variable $\alpha(x)$ replace $\alpha(x)u$ by $\int^u \alpha(x, s) ds$.)

2. (*Collapse*) For some α as above, the **residual** $\tilde{v} := v - \alpha u$ is independent of u on E (i.e., a function of x only).
3. (*Exactness*) The 1-form $\omega = dv - \alpha du$ is **exact** on E : $\omega = d\psi$ for some scalar potential $\psi(x)$ (no u dependence).

Proof. (1) $\Rightarrow$ (2) is immediate: $\tilde{v} = \log \kappa(x)$. (2) $\Rightarrow$ (3): if $\tilde{v} = \psi(x)$, then $d\tilde{v} = dv - \alpha du = d\psi(x)$. (3) $\Rightarrow$ (1): exactness and simple connectivity imply $\tilde{v} = \psi(x) + C$, hence $v = \alpha u + \log \kappa(x)$.

Corollary 3.4 (Flatness test).

On simply connected E , collapse holds iff $d\omega = 0$. In local coordinates,

$$d\omega = -d\alpha \wedge du.$$

Thus a **necessary and sufficient** condition for collapse is that $\partial\alpha/\partial u = 0$ and that any x dependence of α does not create holonomy around loops with u extent. For constant α , $d\omega = 0$ automatically.

Remark. If $\alpha = \alpha(x)$ only, $d\omega = -(\partial\alpha/\partial x) dx \wedge du$. Flatness then requires that along any loop in E with nonzero u extent, the x variation integrates to zero, equivalently, that the field be **path-independent** after gauge fixing. In practice we work on small bins where α is approximately constant, so $d\omega \approx 0$.

3.4 Holonomy, regime mixing, and why collapse can (and should) fail

Definition 3.5 (Holonomy of the RTM connection).

Given a closed loop $\gamma \subset E$, define the holonomy

$$\mathcal{H}(\gamma) = \oint_{\gamma} \omega = \oint_{\gamma} (dv - \alpha du).$$

- If $\mathcal{H}(\gamma) = 0$ for all loops (i.e., $d\omega = 0$), residuals are path-independent and collapse can succeed.
- If $\mathcal{H}(\gamma) \neq 0$ for some loop, the bin contains **incompatible regimes** or genuine curvature (non-power behavior): collapse must fail.

Proposition 3.6 (Mixtures and curvature induce holonomy).

Suppose E contains subregions with different exponents $\alpha_1 \neq \alpha_2$ across a seam in u or x . Any loop encircling the seam yields $\mathcal{H}(\gamma) \neq 0$. Hence collapse across the whole E is impossible; the set must be rebinned.

Sketch. Integrate ω piecewise; the jump in α contributes a nonzero integral of $(\alpha_2 - \alpha_1) du$.

3.5 Variable-exponent case and adiabatic collapse

Empirically, α may drift slowly with u or x . Then exact collapse cannot hold globally, but **adiabatic collapse** can hold on short windows.

Proposition 3.7 (Adiabatic approximation).

If $\alpha(x, u)$ is C^1 and $\|\partial\alpha/\partial u\| \leq \varepsilon$ on E , then over any u window of width h ,

$$\tilde{v}(x, u) = v - \alpha(u_0, x)u = \log\kappa(x) + O(\varepsilon h),$$

uniformly for $u \in [u_0 - h/2, u_0 + h/2]$. Consequently, the empirical collapse statistic $R^2(\tilde{v} \sim u)$ is $O(\varepsilon^2 h^2)$.

Sketch. First-order Taylor expansion of $\alpha(u)$ around u_0 ; bound the residual trend.

Interpretation. This justifies RTM's **binning doctrine**: make windows small enough that curvature is negligible; collapse then tests approximate flatness.

3.6 Identifiability under gauge (global view)

Proposition 3.8 (Gauge equivalence class).

Two time fields v_1, v_2 on E define the same α iff their RTM 1-forms differ by an exact pullback from X :

$$dv_2 - \alpha du = dv_1 - \alpha du + d\phi(x).$$

Equivalently, $v_2 = v_1 + \phi(x)$. Thus the **moduli** of RTM structures on E is the quotient

$$\mathcal{M}(E) \cong \{(\alpha, v)\} / \{v \sim v + \phi(x)\}.$$

Slope α classifies the orbit; clocks live in the gauge fiber.

Consequence. Any empirical procedure that estimates α from slopes in u is automatically gauge-invariant; procedures that use levels in v are not.

3.7 Practical diagnostics in geometric language

- **Collapse statistic** $\Delta_{\text{collapse}} = R^2(\tilde{v} \sim u)$ is a **curvature proxy**; large values indicate $d\omega \neq 0$ or regime mixing.
- **Clock placebos** (changing time units) implement $v \mapsto v + \text{const}$: they should not change α or Δ_{collapse} .
- **Rebinning** corresponds to restricting to subdomains where $d\omega \approx 0$.
- **Ledger of intercepts** is a record of chosen gauges $\phi(x)$ across datasets.

3.8 Summary

- RTM is naturally expressed via the 1-form $\omega = d \log T - \alpha d \log L$.
- **Clock changes** are gauge transformations $\omega \mapsto \omega + d\phi(x)$; **slope** α is gauge-invariant.
- **Collapse $\Leftrightarrow$ exactness/flatness** of ω on a bin; holonomy/curvature explains when collapse must fail.
- **Adiabatic collapse** holds on small windows when α varies slowly, quantifying finite-window bias.

4. Variable-Exponent Regular Variation (Analysis)

Section 2 derived exact power laws from scale symmetry. Empirically, RTM often holds **locally** while exponents drift **slowly** across environments or across the scale axis. This section places RTM within **regular variation** with **spatially varying** indices, gives representation theorems, and quantifies finite-window bias and the linkage between **collapse statistics** and **curvature**.

Throughout, write $x \in X$ for environment, $L > 0$ for scale, $u = \log L$, and $v(x, u) = \log T(x, L)$.

4.1 Classical regular variation (recap)

A measurable $T: \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ is **regularly varying** of index $\alpha \in \mathbb{R}$ if

$$\lim_{L \rightarrow \infty} \frac{T(bL)}{T(L)} = b^\alpha \forall b > 0.$$

Then (Karamata–de Haan) there exists a **slowly varying** ℓ such that $T(L) = L^\alpha \ell(L)$, with $\ell(bL)/\ell(L) \rightarrow 1$ for each fixed b . On log–log scales,

$$v(u) = \alpha u + \log \ell(e^u), \quad \partial_u v(u) = \alpha + o(1).$$

Hence local slopes converge to α as $u \rightarrow \infty$.

Potter bounds. For every $\epsilon > 0$, there exists U such that for $u \geq U$,

$$|\log \ell(e^{u+h}) - \log \ell(e^u)| \leq \epsilon |h| + o(1), \quad h \text{ bounded},$$

which will control bias on finite windows.

4.2 RTM with variable exponent $\alpha(x)$

We now allow the exponent to vary with environment x (and later slowly with u).

Definition 4.1 (Pointwise regular variation in x) $T(\cdot; x)$ is **regularly varying at ∞** with index $\alpha(x)$ if for each fixed x and $b > 0$,

$$\lim_{L \rightarrow \infty} \frac{T(bL; x)}{T(L; x)} = b^{\alpha(x)}.$$

Equivalently,

$$T(L; x) = L^{\alpha(x)} \ell(L; x),$$

where $\ell(\cdot; x)$ is slowly varying **uniformly on compact sets of x** (UCS): for each compact $K \subset X$ and $b > 0$,

$$\sup_{x \in K} \left| \frac{\ell(bL; x)}{\ell(L; x)} - 1 \right| \xrightarrow{L \rightarrow \infty} 0.$$

Proposition 4.2 (Uniform local slope).

Under UCS slow variation,

$$\partial_u v(x, u) = \alpha(x) + r(x, u), \sup_{x \in K} |r(x, u)| \rightarrow 0 (u \rightarrow \infty)$$

for each compact $K \subset X$. Thus in large-scale bins, **binwise slopes** converge uniformly to $\alpha(x)$.

Sketch. Take logs, differentiate in u ; the UCS property gives uniform smallness of the increment of $\log \ell$.

4.3 Drift in α across scale: $\alpha(x, u)$

Empirically, exponents can **drift slowly with u** (finite-range phenomena, evolving regimes).

Model

$$v(x, u) = \int_{u_0}^u \alpha(x, s) ds + \log \kappa(x, u),$$

with κ slowly varying in the sense that for bounded h ,

$$\sup_{x \in K} |\log \kappa(x, u + h) - \log \kappa(x, u)| \leq \epsilon |h| + o(1),$$

and assume **adiabaticity**:

$$\sup_{x \in K} |\partial_u \alpha(x, u)| \leq \epsilon \quad (\text{small}).$$

Theorem 4.3 (Adiabatic representation and bias bound).

Let $\hat{\alpha}(x; u, h)$ be any **symmetric local slope** estimator on the window $[u - h/2, u + h/2]$ (e.g., ODR/TLS/Theil–Sen). Under the slow-drift and slow-variation conditions,

$$\hat{\alpha}(x; u, h) = \alpha(x, u) + O(\epsilon h) + O(\epsilon).$$

Hence finite-window bias is linear in **curvature** $\partial_u \alpha$ and bounded by the slow variation of κ .

Sketch. Taylor expand $v(x, u + s)$ to first order in s with remainder $\frac{1}{2}(\partial_u \alpha) s^2$; symmetric windows cancel odd terms; Potter bounds handle κ .

Corollary 4.4 (Collapse statistic under slow drift).

Let $\tilde{v}(x, u) = v(x, u) - \hat{\alpha}(x; u, h)u$ within the window. Then

$$R^2(\tilde{v} \sim u) = O((\varepsilon h)^2) + O(\varepsilon^2),$$

i.e., the **collapse failure** scales quadratically with window width and the curvature of α .

4.4 Specification test: curvature vs. power law

Suppose the true relation is **non-power** with twice differentiable g :

$$v(u) = g(u), g''(u) \neq 0.$$

Let $\hat{\alpha}(u, h)$ be the local least-squares slope on $[u - h/2, u + h/2]$.

Lemma 4.5 (Local linearization error).

$$\sup_{|s| \leq h/2} |g(u + s) - (g(u) + \hat{\alpha}(u, h)s)| \geq c|g''(u)|h^2$$

for some universal constant $c > 0$. Consequently,

$$R^2(\tilde{v} \sim u) \geq c' |g''(u)|^2 h^2 + o(h^2),$$

so persistent curvature forces **non-vanishing collapse statistic** as $h \rightarrow 0$ only linearly, giving a practical **specification test** against power laws.

Sketch. Chebyshev alternation / Taylor remainder bounds; the regression residual variance lower-bounded by curvature energy.

4.5 Multi-regime mixtures and identifiability

Let $v(u) = \alpha_1 u + c_1$ on $[u_-, u^*]$ and $v(u) = \alpha_2 u + c_2$ on $[u^*, u_+]$ with $\alpha_1 \neq \alpha_2$.

Proposition 4.6 (Unavoidable holonomy / collapse failure).

Any single-window slope over $[u_-, u_+]$ exhibits residual trend with magnitude $\Omega(|\alpha_2 - \alpha_1| |u_+ - u_-|)$. Thus **rebins** that respect regime boundaries are necessary; otherwise the geometry of Sec. 3 yields nonzero holonomy.

Sketch. Piecewise linear exactness implies kink at u^* ; any single affine fit leaves systematic sign-changing residuals.

4.6 Errors-in-variables under regular variation

Let $x = \log L$, $y = v(x) = \alpha(x_0)x + c + r(x)$ in a fixed bin around x_0 , with $|r'(x)| \leq \varepsilon$ (small curvature). Observations satisfy

$$x^{obs} = x + \xi, y^{obs} = y + \zeta,$$

with mean-zero errors, ξ independent of x , and ζ independent noise.

Theorem 4.7 (Consistency of ODR/SIMEX with slow drift).

If $\mathbb{E}\xi^2 < \infty$, $\mathbb{E}\zeta^2 < \infty$, and curvature $\varepsilon \rightarrow 0$ with window width, then:

- **ODR/TLS** slope $\hat{\alpha}_{ODR} \rightarrow \alpha(x_0)$ in probability as $n \rightarrow \infty$, window $h \rightarrow 0$ with $nh \rightarrow \infty$.

- **SIMEX** is consistent provided an accurate estimate of $\text{Var}(\xi)$ is available; the SIMEX extrapolation error is $o(1)$ under the same regime.

Sketch. Standard EIV asymptotics on local linear models plus bias control from Theorem 4.3.

4.7 Putting it together (operational rules)

1. **Binning doctrine.** Choose windows small enough that $\partial_u \alpha$ is negligible: bias = $O(\varepsilon h)$.
2. **Collapse as curvature proxy.** Use $R^2(\tilde{v} \sim u)$ to detect curvature g'' or regime mixtures; thresholds scale like h^2 .
3. **Clock robustness.** Slowly varying clocks κ alter intercepts, not slopes; Potter bounds control their contribution.
4. **EIV aware.** Use ODR/SIMEX/Theil–Sen; ensure $nh \rightarrow \infty$ for consistency while $h \rightarrow 0$ for bias control.
5. **Mixture detection.** Kinks (piecewise α) imply nonzero holonomy (Sec. 3) and force rebinning.

4.8 Summary

- RTM fits naturally inside **regular variation**: exact power when α is constant; **locally power-like** when α drifts slowly.
- Finite-window **slope bias** is $O(\partial_u \alpha \cdot h)$; collapse failures scale like $O((\partial_u \alpha)^2 h^2)$.
- **Curvature** or **mixtures** produce persistent collapse residuals; this yields a principled **specification test** and justifies binning.
- With measurement error, **ODR/SIMEX** remain consistent for the local exponent under standard regimes.

5. Renormalization on Scales: Fixed Points and Stability

RTM's power law emerges from scale symmetry. We now recast this as a **renormalization** problem on a function space: rescale the argument $L \mapsto bL$ and re-gauge the clock so the result can be compared at the original scale. **Power laws are fixed points** of this operator; under mild contractivity, flows approach the power-law manifold, giving a dynamical justification for RTM.

5.1 Function spaces and the renormalization operator

Let $\mathcal{F}$ be a class of positive functions $T: \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ with $\log T \in C_{\text{loc}}^1$. Fix $b > 1$ (a dilation). For a given **gauge choice** $f(b) > 0$ (the clock factor), define:

$$(\mathcal{R}_b T)(L) := \frac{T(bL)}{f(b)}.$$

Typical gauges:

- **Exact-slope gauge** when α is known: $f(b) = b^\alpha$.
- **Self-normalizing gauge**: $f(b) = T(bL_0)/T(L_0)$ for a reference $L_0 > 0$ (so $(\mathcal{R}_b T)(L_0) = T(L_0)$).
- **Moment gauge**: choose $f(b)$ so a chosen functional $\Phi[T]$ is invariant (e.g., $\Phi[T] = \int w(L) \log T(L) dL$).

We will work primarily with the **self-normalizing gauge**; results carry to other gauges by equivalence (Remark 5.2).

Metric. On compact $I \subset (0, \infty)$, use

$$d_I(T_1, T_2) = \sup_{L \in I} |\log T_1(L) - \log T_2(L)|.$$

On $\mathcal{F}$, consider the projective family $d = \sum_{k=1}^{\infty} 2^{-k} d_{I_k}$ for nested compacts $I_k = [e^{-k}, e^k]$.

5.2 Fixed points are power laws

Proposition 5.1 (Fixed points).

Let $f(b) = b^\alpha$ for some $\alpha \in \mathbb{R}$. Then T is a fixed point of $\mathcal{R}_b$ for all $b > 1$,

$$\mathcal{R}_b T = T \forall b > 1,$$

iff $T(L) = \kappa L^\alpha$ for some $\kappa > 0$.

Proof. If $T(L) = \kappa L^\alpha$, then $T(bL)/b^\alpha = \kappa(bL)^\alpha/b^\alpha = \kappa L^\alpha = T(L)$. Conversely, assume $\mathcal{R}_b T = T$ for all b . Then $T(bL) = b^\alpha T(L)$; by Theorem 2.5, $T(L) = \kappa L^\alpha$.

Remark 5.2 (Gauge equivalence).

With the self-normalizing gauge $f(b) = T(bL_0)/T(L_0)$, fixed points satisfy $T(bL)/T(bL_0) = T(L)/T(L_0)$, i.e.,

$$\frac{T(bL)}{T(L)} = \frac{T(bL_0)}{T(L_0)} = b^\alpha,$$

so fixed points are again precisely power laws. Thus **fixed points are gauge-invariant up to clock**.

5.3 Linearization and stability near a power law

Let $T^*(L) = \kappa L^\alpha$ be a fixed point (for gauge $f(b) = b^\alpha$). Write perturbations in log-space:

$$\log T(L) = \log T^*(L) + \varepsilon(L), \varepsilon: \mathbb{R}_{>0} \rightarrow \mathbb{R}.$$

Then

$$\log(\mathcal{R}_b T)(L) = \log T^*(L) + \varepsilon(bL) - \log f(b) + \log \kappa + \alpha \log L - (\log \kappa + \alpha \log L).$$

Hence, for the exact-slope gauge ($f(b) = b^\alpha$):

$$\varepsilon \mapsto \mathcal{L}_b \varepsilon \quad \text{with } (\mathcal{L}_b \varepsilon)(L) = \varepsilon(bL).$$

Thus the linearized renormalization acts by **composition with dilation**.

Lemma 5.3 (Contraction on Hölder/Zygmund classes).

Let $\mathcal{C}^{0,\beta}$ be Hölder functions of $u = \log L$ with seminorm $[\varepsilon]_\beta = \sup_{u \neq v} \frac{|\varepsilon(e^u) - \varepsilon(e^v)|}{|u - v|^\beta}$. Then for any $b > 1$,

$$[\mathcal{L}_b \varepsilon]_\beta = b^{-\beta} [\varepsilon]_\beta.$$

If we use the norm $\| \varepsilon \|_{C^{0,\beta}(I)} = \sup_I | \varepsilon | + \text{diam}(I)^\beta [\varepsilon]_\beta$ on a compact interval I stable under $u \mapsto u + \log b$, the operator is a **strict contraction** with factor $b^{-\beta} < 1$.

Proof. In u coordinates, $(\mathcal{L}_b \varepsilon)(e^u) = \varepsilon(e^{u+\log b})$; differences shrink by $b^{-\beta}$.

Theorem 5.4 (Local stability of power laws).

Fix a compact $I \subset \mathbb{R}$ (in $u = \log L$), and let the gauge be $f(b) = b^\alpha$. If $\varepsilon \in C^{0,\beta}$ on $I' = \{u + \log b^n : u \in I, n = 0, 1, 2, \dots\}$ with small norm, then the iterates satisfy

$$\| \varepsilon_n \|_{C^{0,\beta}(I)} \leq b^{-n\beta} \| \varepsilon_0 \|_{C^{0,\beta}(I')} \xrightarrow{n \rightarrow \infty} 0,$$

i.e., $\mathcal{R}_b^n T \rightarrow T^*$ uniformly on I in log-space. Hence **power laws are locally attractive** in Hölder/Zygmund topologies.

Interpretation. Small Hölder perturbations are **damped** by repeated rescaling (they get “shifted right” in u and smoothed). This is the dynamical counterpart of **regular variation**.

5.4 Slowly varying clocks and center manifolds

Let $T(L) = L^\alpha \kappa(L)$ with **slowly varying** κ . In log-space: $\varepsilon(u) = \log \kappa(e^u)$ with $\varepsilon(u+h) - \varepsilon(u) \rightarrow 0$ as $u \rightarrow \infty$.

Proposition 5.5 (Center manifold of slowly varying factors).

Under the self-normalizing gauge $f(b) = T(bL_0)/T(L_0)$, the renormalization dynamics on ε is

$$(\mathcal{L}_b \varepsilon)(u) = \varepsilon(u + \log b) - \varepsilon(u_0 + \log b) + \varepsilon(u_0),$$

which preserves the “anchor” $\varepsilon(u_0)$ and shifts **differences** along u . If ε is slowly varying, then for any compact I ,

$$\sup_{u \in I} |(\mathcal{L}_b^n \varepsilon)(u) - \varepsilon(u_0)| \xrightarrow{n \rightarrow \infty} 0.$$

Thus $L^\alpha \kappa(L)$ flows to the **power-law leaf** determined by the chosen gauge; the slowly varying factor sits on a **center manifold** (neutral direction) that is quotiented out by self-normalization.

Sketch. Telescoping sums of slow increments; uniform convergence on compacts follows from Potter bounds.

5.5 Variable exponents and adiabatic attraction

Let $T(L) = \exp(\int_{u_0}^{\log L} \alpha(s) ds) \kappa(L)$ with $\alpha \in C^1$ and $|\alpha'(u)| \leq \varepsilon$ small on a band encompassing $I' = \cup_{n \geq 0} (I + n \log b)$.

Theorem 5.6 (Adiabatic stability to a drifting power law).

Under the exact-slope gauge $f(b) = b^{\alpha(u_0)}$ or the self-normalizing gauge, the iterates satisfy on any compact I :

$$\sup_{u \in I} | \log(\mathcal{R}_b^n T)(e^u) - (\alpha(u_0) u + C_n) | \leq C \varepsilon n \log b + o(1),$$

where C_n is a (gauge-dependent) constant. For fixed I , as n grows the right-hand side remains **small** provided the cumulative drift $\varepsilon n \log b$ is small, this is the **adiabatic regime**. Hence on finite windows the flow **tracks** a local power law with exponent near $\alpha(u_0)$.

Sketch. Decompose $\int_{u_0}^{u+n\log b} \alpha(s) ds = \alpha(u_0)(u + n\log b - u_0) + \int \alpha'(s)(u + n\log b - s) ds$. The remainder scales with $\varepsilon n \log b$; slow variation of κ handled as in 5.5.

Interpretation. If α drifts slowly, renormalization still pushes toward **local power-law behavior** on any fixed window, precisely the empirical RTM setting.

5.6 Non-power alternatives: curvature generates unstable modes

Consider $v(u) = g(u)$ with curvature $g'' \not\equiv 0$. In perturbation coordinates relative to α , set $\varepsilon(u) = g(u) - (\alpha u + c)$. Then under $\mathcal{L}_b$,

$$\varepsilon(u) \mapsto \varepsilon(u + \log b).$$

If g'' is persistent (e.g., periodic or polynomial), the shifted residuals **do not decay** on fixed windows; they only “translate.” The contraction of Lemma 5.3 fails in C^0 unless we quotient out by drift and curvature. Consequently:

Proposition 5.7 (Curvature as a non-decaying mode).

If g'' does not vanish at infinity (or decays too slowly), then for any gauge, there exists a compact window I and $\delta > 0$ such that

$$\inf_{n \geq 0} \sup_{u \in I} |\varepsilon_n(u)| \geq \delta,$$

i.e., **renormalization does not contract** to a power law on that window. This aligns with **collapse failure** (Sec. 4.4).

Conclusion. Persistent curvature is an **unstable feature** under RG flow, precisely what our collapse test detects.

5.7 Summary and implications

- The renormalization operator $\mathcal{R}_b$ formalizes “**zooming out in scale and re-gauging the clock**”.
- **Power laws are fixed points**, independent of gauge (up to a multiplicative constant).
- In Hölder/Zygmund topologies, $\mathcal{R}_b$ is a **contraction**, giving **local attraction** to the power-law manifold.
- **Slowly varying clocks** lie on a **center manifold** and are neutralized by self-normalization; flows converge to a representative power law.
- **Slowly drifting exponents** yield **adiabatic attraction**: on any fixed window, iterates track a local power law with small, controlled error.
- **Curvature** in $g = \log T$ is a **non-decaying mode**; under RG it persists as translation, exactly mirroring **collapse failure**.

6. Scale-Dependent Diffusions and Dirichlet Forms

This section shows how an RTM exponent acts as a **local clock field** in stochastic/PDE dynamics. We construct diffusions on metric spaces whose *effective time* stretches with scale, derive self-similar laws when α is constant, and prove **adiabatic** approximations when α varies slowly. This links RTM to sub/super-diffusion *without* committing a priori to fractional operators.

6.1 Metric–measure setting and RTM conductivity

Let (M, d, μ) be a complete, separable metric measure space with a regular, strongly local Dirichlet form $(\mathcal{E}, \mathcal{D})$ on $L^2(\mu)$ and carré-du-champ Γ . For intuition, $M = \mathbb{R}^m$ with $\Gamma(u) = |\nabla u|^2$.

Let $L: M \rightarrow (0, \infty)$ be a **scale proxy** (e.g., local neighborhood radius, degree, or coarse density) and $\alpha: M \rightarrow \mathbb{R}$ a **coherence field**. Define an **RTM conductivity**

$$D(x) = L(x)^{-\alpha(x)} \quad (\text{slower clocks at larger scale if } \alpha > 0).$$

Definition 6.1 (RTM Dirichlet form).

$$\mathcal{E}_\alpha(u, v) = \int_M D(x) \Gamma(u, v)(x) d\mu(x), \mathcal{D}_\alpha = \mathcal{D}.$$

This is closed, symmetric, and generates a conservative Markov semigroup $(P_t^\alpha)_{t \geq 0}$ with generator

$$\mathcal{L}_\alpha u = \nabla \cdot (D \nabla u) \text{ (in } \mathbb{R}^m \text{ case).}$$

6.2 Constant exponent: self-similar scaling

Assume $\alpha(x) \equiv \alpha$ and $L(x) = \lambda |x|_R$ for some homogeneous quasi-norm $|\cdot|_R$ of degree 1 under a dilation group $x \mapsto b x$.

Theorem 6.2 (RTM similarity).

Let $u(t, x)$ solve $\partial_t u = \mathcal{L}_\alpha u$ with integrable initial data. Then for any $b > 0$,

$$u(t, x) = b^m u(b^{m+\alpha} t, b x)$$

in the sense of distributions. In particular, the heat kernel $p^\alpha(t, x, y)$ obeys

$$p^\alpha(t, x, y) = t^{-\frac{m}{m+\alpha}} \Phi \left(\frac{d(x, y)}{t^{1/(m+\alpha)}} \right)$$

for some profile Φ (Gaussian-like tails when $M = \mathbb{R}^m$). Thus the **diffusion radius** scales as

$$r(t) \asymp t^{1/(m+\alpha)} \Leftrightarrow t \asymp r^{m+\alpha}.$$

Interpretation. The **effective dynamic exponent** is $z = m + \alpha$: time grows with scale as $T \sim L^z$. When m is fixed, varying α changes the **clock gradient** across scale.

Sketch. Invariance of $\mathcal{E}_\alpha$ under $x \mapsto b x, t \mapsto b^{m+\alpha} t$, and mass conservation give the scaling law.

6.3 Slowly varying exponent: adiabatic clocks

Let $\alpha \in C^1(M)$ and $L \in C^1(M)$. Consider the inhomogeneous PDE

$$\partial_t u = \nabla \cdot (L(x)^{-\alpha(x)} \nabla u).$$

Assumption (adiabatic drift). There exists $\varepsilon \ll 1$ and a covering of M by patches U_k of diameter h such that

$$\sup_{x \in U_k} \|\nabla \alpha(x)\| \leq \varepsilon, \quad \sup_{x \in U_k} \|\nabla \log L(x)\| \leq \varepsilon.$$

Theorem 6.3 (Local self-similarity, adiabatic error).

Fix a patch U and a reference point $x_0 \in U$. Let $\alpha_0 = \alpha(x_0)$, $L_0 = L(x_0)$. For times t such that the diffusion radius $r(t) \ll h$,

$$u(t, x) = (P_t^{\alpha_0} u_0)(x) + \mathcal{O}(\varepsilon t r(t)^{-2}) \quad \text{uniformly for } x \in U,$$

with $r(t) \asymp t^{1/(m+\alpha_0)}$. Equivalently, on **finite observation windows**, the solution is approximated by a constant- α_0 model with an **adiabatic error** linear in the local curvature of the clock field.

Sketch. Duhamel expansion around the frozen-coefficient operator at x_0 ; commutator bounds yield the stated error using local Gaussian estimates.

Corollary 6.4 (Adiabatic ECI consistency).

Local estimates of the **time-scale slope** from solution features (e.g., heat-ball radius vs. time, first-passage times) converge to $\alpha(x_0)$ with bias $\mathcal{O}(\varepsilon)$ as the observation window shrinks, matching the statistical bias bounds of Section 4.

6.4 First-passage and exit times

Let $\tau_R = \inf\{t > 0: X_t \notin B(x_0, R)\}$ for the diffusion generated by $\mathcal{L}_\alpha$.

Proposition 6.5 (RTM exit-time scaling).

If $\alpha \equiv \alpha_0$ and $L(x) \propto |x - x_0|$ near x_0 , then

$$\mathbb{E}_{x_0} \tau_R \asymp R^{m+\alpha_0}.$$

Under adiabatic drift, for $R \ll h$,

$$\mathbb{E}_{x_0} \tau_R = c(x_0) R^{m+\alpha(x_0)} (1 + \mathcal{O}(\varepsilon R)).$$

Interpretation. Exit-time exponents **directly encode** the local RTM slope. These are operational observables for empirical α in spatial systems.

6.5 Sub-/super-diffusion without fractional Laplacians

Classically, anomalous diffusion uses fractional operators $(-\Delta)^\beta$. Here, RTM achieves **effective anomaly** via **space-dependent clocks** instead:

- If $\alpha > 0$, large scales are **slower** (sub-diffusive w.r.t. Euclidean time).
- If $\alpha < 0$, large scales are **faster** (super-diffusive).

Proposition 6.6 (Equivalence in self-similar class).

In $\mathbb{R}^m$ with $\alpha \equiv \alpha_0$, the heat kernel has the same similarity exponent as a fractional diffusion with order $\beta = \frac{2}{m+\alpha_0}$ in the sense that the mean squared displacement satisfies

$$\mathbb{E} |X_t|^2 \asymp t^{2/(m+\alpha_0)} = t^\beta.$$

Hence RTM's constant- α family **realizes** sub/super-diffusion exponents via local clock modulation rather than nonlocal jumps.

6.6 Spectral viewpoint

Let $\{-\mathcal{L}_\alpha \varphi_k = \lambda_k \varphi_k\}$ be the spectral resolution (on a bounded domain with Dirichlet boundary).

Theorem 6.7 (Weyl-type law with RTM clock).

If L, α are smooth and bounded above/below on $\Omega \subset \mathbb{R}^m$,

$$N(\lambda) := \#\{k: \lambda_k \leq \lambda\} \sim C_m \int_{\Omega} (\lambda L(x)^{\alpha(x)})^{m/2} dx (\lambda \rightarrow \infty).$$

Consequently, high-frequency eigenmodes “feel” the **local clock** as a density multiplier L^α .

Sketch. Local Weyl law via semiclassical measure with variable coefficient principal symbol $|\xi|^2 L(x)^{-\alpha(x)}$.

6.7 Stochastic representation and time change

Let B_t be Brownian motion on M (for $M = \mathbb{R}^m$). Define an **additive functional**

$$A_t = \int_0^t D(B_s) ds = \int_0^t L(B_s)^{-\alpha(B_s)} ds,$$

and its right-continuous inverse $T(t) = \inf\{s: A_s > t\}$.

Proposition 6.8 (Clock-change representation).

The diffusion $X_t = B_{T(t)}$ has generator $\mathcal{L}_\alpha$. Thus RTM diffusions are **time-changed Brownian motions** with a *state-dependent clock*.

Consequences. Many properties (martingales, Harnack bounds) lift from Brownian motion through the time change, clarifying when RTM inherits classical regularity.

6.8 Summary

- RTM enters diffusion theory as a **space-dependent clock** $L^{-\alpha}$ multiplying conductivity.
- **Constant** α yields **exact similarity** with dynamic exponent $z = m + \alpha$ and exit times $T \sim R^z$.
- **Slowly varying** α admits **adiabatic** approximations; local estimates of the time-scale slope are consistent with controlled bias.
- RTM provides an alternative path to **anomalous diffusion** and a clean **spectral** interpretation; RTM diffusions are **time-changed Brownian motions**.

7. Identifiability & Statistical Consistency

This section formalizes **what is identifiable** from finite, noisy data and gives **consistency** results for common slope estimators used in RTM: **orthogonal distance regression (ODR/TLS)**, **SIMEX**, and **Theil-Sen**. We also cast the **collapse statistic** as a specification test against non-power alternatives with measurement error.

Setup: in a fixed bin (environment), the ideal model is

$$y = \log T = \alpha x + c + r(x), \quad x = \log L,$$

where $r \equiv 0$ (exact RTM) or $|r'(x)| \leq \varepsilon$ (slow drift / curvature). Observations are noisy:

$$x^{obs} = x + \xi, \quad y^{obs} = y + \zeta,$$

with $\mathbb{E}[\xi] = \mathbb{E}[\zeta] = 0$, finite variances, and mild independence/regularity given below.

7.1 What is (and is not) identifiable

Proposition 7.1 (Slope is clock-invariant; intercept is not).

If the clock rescales as $y^\# = y + \phi$ with ϕ constant in x inside the bin, then any slope estimator based on contrasts in x (ODR, SIMEX, Theil–Sen) is unchanged, while the intercept shifts by ϕ .

Implication. Only α is an intrinsic target; intercepts are gauge (clock) artifacts.

Proposition 7.2 (Identifiability up to curvature).

If $r \equiv 0$, α is point-identified from the joint distribution of (x^{obs}, y^{obs}) given measurement error structure. If $|r'| \leq \varepsilon$, then the identified target is the **local slope** $\alpha(u_0)$ up to bias $O(\varepsilon h)$ for window width h (Section 4).

7.2 Orthogonal Distance Regression (Total Least Squares)

Assume i.i.d. sample $\{(x_i^{obs}, y_i^{obs})\}_{i=1}^n$, with

- (i) x_i supported on a compact interval $[a, b]$, density bounded away from 0;
- (ii) ξ_i, ζ_i independent of x_i and of each other, mean 0, finite second moments;
- (iii) $r \equiv 0$ (or $|r'| \leq \varepsilon$ small on window).

Theorem 7.3 (Consistency of ODR/TLS).

Under (i)–(ii) with $r \equiv 0$, the ODR slope $\hat{\alpha}_{ODR}$ is **consistent** for α , and $\sqrt{n}(\hat{\alpha}_{ODR} - \alpha)$ is asymptotically normal with variance determined by the second moments of (x, ξ, ζ) . If moreover $|r'| \leq \varepsilon$ on a window of width h , then

$$\hat{\alpha}_{ODR} = \alpha(u_0) + O_p(\varepsilon h) + O_p(n^{-1/2}).$$

Sketch. Classical TLS asymptotics (eigenvector of centered covariance matrix). Curvature contributes a deterministic bias of order εh .

Remark. OLS is **attenuated** when $\xi \neq 0$; ODR is the correct EIV remedy when the error ratio is not extreme.

7.3 SIMEX (Simulation–Extrapolation)

Assume we know or can estimate $\sigma_\xi^2 = \text{Var}(\xi)$. Define pseudo-samples

$$x_i^{(\lambda)} = x_i^{obs} + \sqrt{\lambda} \xi_i, \lambda \in \Lambda \subset [0, \Lambda_{\max}],$$

with fresh $\xi_i \sim N(0, \sigma_\xi^2)$; fit naive slopes $\hat{\alpha}(\lambda)$ (e.g., OLS or ODR) and extrapolate a low-order polynomial to $\lambda = -1$ to obtain $\hat{\alpha}_{SIMEX}$.

Theorem 7.4 (Consistency of SIMEX).

If σ_ξ^2 is consistently estimated and $r \equiv 0$, then $\hat{\alpha}_{SIMEX} \xrightarrow{p} \alpha$. With $|r'| \leq \varepsilon$, in a window h ,

$$\hat{\alpha}_{SIMEX} = \alpha(u_0) + O_p(\varepsilon h) + o_p(1).$$

Sketch. Standard SIMEX theory: the measurement error bias is a smooth function of λ ; extrapolating to -1 removes it.

Practical note. When σ_ξ^2 is misspecified, SIMEX may over/under-correct; use as a **sensitivity bound** alongside ODR.

7.4 Theil–Sen (Robust median slope)

Define the median of pairwise slopes on (x^{obs}, y^{obs}) . Under symmetric noise and no curvature, Theil–Sen is $\sqrt{n}$ -**consistent** and robust to outliers.

Proposition 7.5 (Robustness envelope).

If a fraction $\pi < 0.29$ of observations are arbitrary outliers, Theil–Sen’s slope still converges to α (breakdown $\sim 29\%$). With small curvature $|r'| \leq \varepsilon$, bias is $O(\varepsilon h)$.

Use. Report Theil–Sen as a **robust check**; fuse with ODR via meta-analysis to guard against heavy tails.

7.5 Inference: uncertainty and small-sample practice

- **Cluster/bootstrapped** by entity or track to capture serial dependence in x and T .
- **Orthogonal residual bootstrap** is appropriate for ODR; **pairs bootstrap** for Theil–Sen.
- Report **percentile CIs** and **influence diagnostics** (leverage points in x).
- Maintain a **slope–intercept ledger**: slopes with CIs, intercepts (gauge), and known clock/unit shifts.

7.6 Collapse statistic with measurement error

Let $\tilde{y}_i = y_i^{obs} - \hat{\alpha} x_i^{obs}$ within a bin, and regress $\tilde{y}$ on x^{obs} . Define

$$\Delta_{\text{collapse}} := R^2(\tilde{y} \sim x^{obs}).$$

Theorem 7.6 (Specification test under EIV).

Assume $r \equiv 0$ (true power law), $\hat{\alpha}$ is consistent, and ξ, ζ are mean-zero with finite variances.

Then $\Delta_{\text{collapse}} \xrightarrow{p} 0$.

If $v = g(x)$ with $g'' \neq 0$ on the bin and mild smoothness, then for any consistent slope estimator,

$$\liminf_{n \rightarrow \infty} \Delta_{\text{collapse}} \geq c \mathbb{E}[g''(X)^2] h^2 (\text{up to error-variance terms}),$$

so the statistic stays **bounded away from 0** in the limit as long as curvature persists over window width h .

Sketch. Under the null, residuals are mean-independent of x ; with curvature, regression captures a non-vanishing linear component proportional to second derivatives (Section 4.4), with EIV inflating variance but not erasing the drift.

Practice. Pre-register a collapse threshold (e.g., < 0.05); accompany with **residual vs. x** plots (nonparametric smooth) and a **clock-placebo** check.

7.7 Window selection and changepoints

- **Bias-variance trade-off:** Choose window width h so that $nh \rightarrow \infty$ (variance $\downarrow$) while $h \rightarrow 0$ (bias $O(\varepsilon h)$ $\downarrow$).
- **Changepoints:** Use PELT/Bai-Perron on (x^{obs}, y^{obs}) pairs or on preliminary residuals to avoid mixing regimes (kinks violate collapse).
- **Coverage gates:** Reject bins with too few effective x spans (thin leverage $\rightarrow$ unstable slope).

7.8 Multi-proxy fusion with uncertainty

Given family-wise estimates $(\hat{\alpha}_f, \hat{\sigma}_f^2)$ that passed collapse, apply **random-effects** fusion:

$$\hat{\alpha}_{RE} = \frac{\sum_f w_f \hat{\alpha}_f}{\sum_f w_f}, w_f = \frac{1}{\hat{\sigma}_f^2 + \hat{\tau}^2}, \hat{\tau}^2 = \max \left\{ \frac{Q - (F - 1)}{\sum w_f - \sum w_f^2 / \sum w_f}, 0 \right\}.$$

Report Q , $\hat{\tau}^2$, and **leave-one-family-out** influence. High $\hat{\tau}^2 \Rightarrow$ publish family-wise $\hat{\alpha}_f$ instead of a single number.

7.9 Finite-sample red flags (practical diagnostics)

- **Attenuation creep:** OLS slope $\ll$ ODR slope.
- **Leverage scarcity:** Most leverage from extreme x points; run **jackknife** by removing them.
- **High Δ_{collapse} :** residual trend vs. $x \rightarrow$ likely curvature or regime mix.
- **Clock failure:** Unit/clock change alters slope $\rightarrow$ rebin; the slope must be **clock-invariant**.

7.10 Summary

- In a bin, α is the only gauge-invariant estimand.
- **ODR/TLS** and **SIMEX** are consistent for α under standard EIV assumptions; **Theil-Sen** is a robust check.
- Finite-window bias from drift/curvature is $O(\varepsilon h)$; manage with binning and changepoints.
- The **collapse statistic** is a specification test: it tends to 0 under the RTM model and stays positive with non-power curvature, even with measurement error.
- Publish **uncertainty, collapse diagnostics, and heterogeneity**; when fusion heterogeneity is high, prefer family-wise slopes over a single index.

8. A Category-Theoretic Packaging of RTM

This section formalizes RTM as a **gauge theory on a scale-clock bundle**. The goal is not abstraction for its own sake, but a clean language for invariants (slope), gauges (clocks), and functoriality under changes of variables, coarse-graining, and product constructions.

8.1 The category RTM

An object of **RTM** is a triple $A = (X, L, v)$ where:

- X is a (second countable) topological space of **environments**;
- $L: X \rightarrow \mathbb{R}_{>0}$ is a continuous **scale** map (or a trivial factor with coordinate $u = \log L$);
- $v: X \rightarrow \mathbb{R}$ is a continuous **clock potential** $v = \log T$.

Associated to A is the **RTM 1-form**

$$\omega_A = dv - \alpha d(\log L),$$

for some α (constant or a field on X), defined up to **gauge**: $v \sim v + \phi$ with $\phi: X \rightarrow \mathbb{R}$.

A **morphism** $\Phi: A \rightarrow B$ is a pair (φ, ψ) with $\varphi: X \rightarrow Y$ continuous and $\psi: Y \rightarrow \mathbb{R}$ such that

$$\Phi^* \omega_B = \omega_A + d(\psi \circ \varphi),$$

i.e., Φ pulls back the target's 1-form to the source's 1-form **up to gauge**. Composition is $(\varphi_2, \psi_2) \circ (\varphi_1, \psi_1) = (\varphi_2 \circ \varphi_1, \psi_1 + \psi_2 \circ \varphi_1)$.

Interpretation. Different clocks correspond to vertical gauge shifts $v \mapsto v + \phi$. Morphisms are **clock-compatible** reparametrizations of environment/scale.

8.2 Gauge group and moduli

The **gauge group** $\mathcal{G}_X = C^0(X, \mathbb{R})$ acts on objects by $v \mapsto v + \phi$. Two objects are **gauge equivalent** if related by this action.

Proposition 8.1 (Slope as a moduli invariant).

If ω_A and ω_B are gauge equivalent on X , then their α fields agree (a.e.). Conversely, equal α fields define the same class in the **moduli space**

$$\mathfrak{M}(X) = \{\text{objects on } X\} / \mathcal{G}_X.$$

Thus $[A] \in \mathfrak{M}(X)$ is uniquely determined by α and the de Rham cohomology class $[\omega_A] \in H^1(X; \mathbb{R})$; in simply connected bins, $[\omega] = 0$ and the class is fully determined by α .

Consequence. In a bin (simply connected), **slope** is the only intrinsic datum; clocks are pure gauge.

8.3 Collapse = trivialization of the RTM bundle

Let $\pi: X \times \mathbb{R} \rightarrow X$ be the trivial line bundle with fiber coordinate $u = \log L$. Consider the connection 1-form

$$\omega = dv - \alpha(x) du.$$

Theorem 8.2 (Collapse $\Leftrightarrow$ flat trivialization).

On a simply connected bin $E \subset X \times \mathbb{R}$, the following are equivalent:

1. There exists a **global section** $s(x) = (x, u)$ and a gauge ϕ such that, in the trivialization with potential $v^\phi = v + \phi$, $v^\phi(x, u) = \alpha(x)u + c(x)$ (RTM chart).

2. The connection ω is **flat** on E : $d\omega = 0$ and its holonomy vanishes.
3. The empirical **collapse** of Section 3 holds on E .

This repackages Section 3 in categorical language: collapse is existence of a trivialization that straightens v into an affine function of u .

8.4 Products, sums, and coarse-graining (monoidal structure)

Define a **monoidal product** $A \otimes B$ for independent subsystems:

$$(X_A, L_A, v_A) \otimes (X_B, L_B, v_B) = (X_A \times X_B, L_A L_B, v_A + v_B),$$

with $\alpha_{\otimes} = \alpha_A + \alpha_B$ if each obeys a power law.

Proposition 8.3 (Additivity under independent composition).

If both factors are in exact RTM form $v_i = \alpha_i u_i + c_i$ with $u = \log(L_A L_B) = u_A + u_B$, then

$$v_{\otimes} = (\alpha_A + \alpha_B)u + (c_A + c_B),$$

so slopes **add** under multiplicative composition of scales. Gauge transforms distribute.

Coarse-graining functor.

Let $\mathcal{C}_b$ map (X, L, v) to $(X, bL, v - \log f_b)$. With a gauge choice f_b (Section 5), $\mathcal{C}_b$ is an **endofunctor** of RTM; power-law objects are its **fixed points**.

8.5 Natural transformations and clock choices

Two gauge choices f_b and g_b for coarse-graining define endofunctors $\mathcal{C}_b^{(f)}$ and $\mathcal{C}_b^{(g)}$. The map

$$\eta_b: \mathcal{C}_b^{(f)} \Rightarrow \mathcal{C}_b^{(g)}, \quad \eta_b(A) = (\text{id}_X, \psi_b), \quad \psi_b = \log f_b - \log g_b$$

is a **natural transformation**, i.e., a functorial gauge shift commuting with morphisms. This encodes the statement “changing the renormalization gauge is just a clock change.”

8.6 Curvature and obstructions (cohomology)

Let $\Omega^1(E)$ be 1-forms on a bin $E \subset X \times \mathbb{R}$. The curvature is

$$\mathcal{F} = d\omega = -d\alpha \wedge du.$$

- If $\mathcal{F} \neq 0$, collapse is **obstructed**; regime mixing or genuine curvature persists.
- If $\mathcal{F} = 0$ but $H^1(E) \neq 0$, collapse may still fail globally due to **holonomy** (nontrivial cohomology class). Local collapse always holds.

Proposition 8.4 (Cohomological obstruction).

Collapse holds globally on E iff $\mathcal{F} = 0$ and $[\omega] = 0$ in $H^1(E)$. Otherwise one can only collapse **locally** (on simply connected charts), consistent with practical binning.

8.7 Observables as functors

An **observable** (e.g., exit-time exponent, diffusion radius exponent) is a functor $\mathcal{O}: \text{RTM} \rightarrow \mathcal{C}$ (sets, groups, numbers) satisfying:

- **Gauge invariance:** $\mathcal{O}(v + \phi) = \mathcal{O}(v)$;

- **Monoidal additivity:** $\mathcal{O}(A \otimes B) = \mathcal{O}(A) + \mathcal{O}(B)$ when defined.

Example. The **slope functor** $\mathcal{S}$ maps $(X, L, v) \mapsto \alpha$ (as a function on X) and is the terminal gauge-invariant observable in bins with $H^1 = 0$.

8.8 Summary

- **RTM** forms a category where objects carry a **gauge 1-form** $\omega = d \log T - \alpha d \log L$.
- **Morphisms** are reparametrizations that preserve ω up to exact forms; the **gauge group** acts by clock shifts.
- **Slope** α is the moduli invariant; **collapse** equals **flat trivialization** (zero curvature and holonomy).
- **Products** and **coarse-graining** are functorial; power laws are fixed points of coarse-graining endofunctors.
- **Cohomology** captures global obstructions to collapse; binning provides simply connected charts where collapse is feasible.

9. Examples, Counterexamples, and Open Problems

We close the mathematical exposition with worked examples that satisfy RTM exactly, controlled counterexamples that **must** fail collapse, and a short list of open problems suggested by the scale-clock framework.

9.1 Exact RTM examples (collapse holds)

Example 9.1 (Pure power law)

Let $T(L) = \kappa L^\alpha$ on $L > 0$. Then $v = \log T = \alpha \log L + \log \kappa$, $\omega = dv - \alpha d(\log L) = 0$.

- **Collapse:** trivial (residual constant).
- **Renormalization:** fixed point of $\mathcal{R}_b$ for all $b > 0$.

Example 9.2 (Slowly varying clock)

Let $T(L) = L^\alpha \ell(L)$ with ℓ slowly varying (Karamata). On any finite $\log L$ window,

$$v(u) = \alpha u + \varepsilon(u), \varepsilon(u+h) - \varepsilon(u) \rightarrow 0.$$

- **Collapse:** holds up to $O(\sup |\varepsilon(u+h) - \varepsilon(u)|)$.
- **RG:** self-normalized $\mathcal{R}_b$ flows to the power-law leaf (Sec. 5.5).

Example 9.3 (Piecewise homogeneous media in PDE)

On $\mathbb{R}^m$, take $\alpha(x) \equiv \alpha_0$ and $L(x) = c |x|$. The heat kernel of $\partial_t u = \nabla \cdot (L(x)^{-\alpha_0} \nabla u)$ satisfies similarity with exponent $z = m + \alpha_0$ (Sec. 6.2).

- **Observable:** exit time $\mathbb{E}\tau_R \asymp R^z$ recovers α_0 .

9.2 Controlled failures (collapse must fail)

Counterexample 9.4 (Regime seam / kink)

$$T(L) = \begin{cases} \kappa_1 L^{\alpha_1}, & L \leq L^*, \\ \kappa_2 L^{\alpha_2}, & L > L^*, \alpha_1 \neq \alpha_2. \end{cases}$$

- **Geometry:** ω is exact on each side, but loops crossing L^* have nonzero holonomy $\oint \omega = (\alpha_2 - \alpha_1) d(\log L)$.
- **Empirics:** residuals show sign change; Δ_{collapse} bounded away from 0 unless we **rebin**.

Counterexample 9.5 (Curved log-log relation)

Let $v(u) = u + \sin u$ so $T(L) = L e^{\sin(\log L)}$.

- **Curvature:** $g''(u) = -\sin u \neq 0 \Rightarrow$ collapse statistic scales like $c h^2$ (Sec. 4.4).
- **RG:** residuals translate under $\mathcal{R}_b$, never contracting on a fixed window (Prop. 5.7).

Counterexample 9.6 (Clock depending on scale)

If a “clock” factor secretly depends on L : $T^\#(L) = c(L) T(L)$ with non-constant c , then

$$v^\#(u) = \alpha u + \log \kappa + \log c(e^u),$$

and $\omega^\# = \omega + d \log c(e^u)$ acquires a du -**component**.

- **Interpretation:** this is **not** a permissible gauge in RTM (clocks must be L independent in-bin). Collapse should and will fail, correctly flagging misspecification.

9.3 Worked composite constructions

Construction 9.7 (Product systems $\rightarrow$ slope additivity)

Let $T_A(L_A) = \kappa_A L_A^{\alpha_A}$, $T_B(L_B) = \kappa_B L_B^{\alpha_B}$. For independent composition with total scale $L = L_A L_B$ and time $T = T_A T_B$:

$$T(L) = \kappa_A \kappa_B L^{\alpha_A + \alpha_B}, \quad \alpha_{\text{total}} = \alpha_A + \alpha_B$$

(Sec. 8.4). This models cascaded stages whose characteristic times multiply.

Construction 9.8 (Adiabatic patching)

Partition the scale axis into windows $\{I_k\}$ where $\|\partial_u \alpha\| \leq \varepsilon$. On each window, fit α_k ; define a **piecewise-adiabatic** model

$$v(u) = \sum_k \mathbf{1}_{u \in I_k} (\alpha_k(u - u_k) + c_k),$$

with continuity constraints at seams.

- **Error:** $O(\varepsilon |I_k|)$ per patch; collapse holds locally, fails globally if α drifts.

9.4 Bridges to other theories

- **Regular variation / Karamata-de Haan.** RTM lives on the **power-law manifold**; slowly varying clocks are the **center manifold** (Sec. 5.5).
- **Renormalization group.** RTM fixed points are RG fixed points; curvature is a non-decaying mode.

- **Time-changed diffusions.** RTM PDEs are Brownian motions with **state-dependent clocks** (Sec. 6.7).
- **Gauge/connection language.** Collapse $\Leftrightarrow$ flat connection; holonomy captures regime mixing (Secs. 3, 8).

9.5 Open problems

1. **Sharp collapse thresholds.** Prove finite-sample, non-asymptotic bounds linking Δ_{collapse} to curvature g'' under EIV, with optimal constants.
2. **Holonomy detection.** Construct statistical tests that distinguish curvature from **topological** obstructions (nontrivial H^1) using loop integrals of residual 1-forms.
3. **Variable-exponent regular variation on graphs.** Let L be degree or path length on a random graph; establish law of large numbers for local $\hat{\alpha}$.
4. **Inverse problems.** From exit-time data $\mathbb{E}_x \tau_R \asymp R^{m+\alpha(x)}$, reconstruct $\alpha(x)$ (Calderón-type uniqueness with scale-dependent coefficients).
5. **Global gauges on non-simply connected bins.** Classify when a global clock exists (vanishing cohomology class of ω), and give constructive algorithms to trivialize if possible.
6. **Beyond power laws.** Characterize the minimal curvature classes g for which RG becomes contracting after **nonlinear** reparametrizations (e.g., log-poly or spline gauges).
7. **Asymptotics under heavy-tailed measurement error.** Extend ODR/SIMEX consistency to α stable noise; quantify robustness envelopes.
8. **Coupled fields.** Analyze PDEs with feedback $L = L(u, x)$ (scale proxy depends on the state), yielding **nonlinear clocks** and potential bifurcations in α .

10. Mathematical Conclusion

We provided a rigorous backbone for RTM:

- From a **scale semigroup** and mild regularity, characteristic time obeys a **power law** $T = \kappa L^\alpha$; the **slope** α is a **clock-invariant** structural quantity (Sec. 2).
- RTM is most naturally expressed via the **1-form** $\omega = d \log T - \alpha d \log L$; **collapse** equals **exactness/flatness**, while regime mixing and curvature appear as **holonomy** (Sec. 3).
- **Regular variation** with (slowly) **variable exponents** explains finite-window estimates and bias; collapse statistics quantify curvature (Sec. 4).
- A **renormalization** operator on functions has **power laws as fixed points** and is **contractive** in Hölder/Zygmund classes; slowly varying clocks form a **center manifold** (Sec. 5).
- In dynamics, RTM exponents act as **local clock fields** for diffusions/PDEs, yielding similarity exponents $z = m + \alpha$ and **adiabatic** approximations when α drifts (Sec. 6).
- Statistically, **ODR/SIMEX/Theil-Sen** consistently recover local α under EIV, and the **collapse statistic** is a specification test against curvature, even with noise (Sec. 7).

- A **categorical** formulation packages invariance, gauges, and coarse-graining functorially (Sec. 8).

The program yields a compact principle: **structure lives in the slope**, clocks live in the gauge. Where bins are stable and collapse holds, RTM gives a falsifiable, transportable description of how **time stretches with scale**. Where collapse fails, RTM provides a **diagnostic** —curvature or regime mixing— not a fudge. The open problems above outline a path to deepen the theory (non-simply connected gauges, inverse problems, graphs, heavy tails) and to connect it with broader analysis and probability.